

Chapter 3: Demand and Supply Analysis

1. DEMAND

Definition:

Demand = Quantity of a commodity consumers are willing and able to purchase at a certain price during a specific time period

Formula:

$$Q_d = f(P)$$

Where: | Symbol | Full Form | Unit | Description | |——|——|——|——| | Qd | Quantity Demanded | units | Amount buyers want to purchase | | P | Price | Rs./unit, \$/unit | Price of the commodity | | f() | Function of | - | Qd depends on P |

Essential Requirements for Demand:

1. **Will** to purchase (desire)
 2. **Power** to purchase (money/income)
 3. **Price** consideration
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2. LAW OF DEMAND

Statement:

“There is an **INVERSE** relationship between price and quantity demanded, other things remaining constant”

Formula Expression:

$$\begin{cases} P \uparrow \Rightarrow Q_d \downarrow \\ P \downarrow \Rightarrow Q_d \uparrow \end{cases}$$

Where: - $\uparrow$ = Increase - $\downarrow$ = Decrease - $\Rightarrow$ = Leads to/Results in

When to Use:

Predicting consumer behavior with price changes
Pricing strategy decisions
Market demand analysis

Example:

Statement: “Bread price increases from \$2 to \$3. Demand falls from 100 to 70 loaves. Verify law of demand.”

Price: \$2 $\rightarrow$ \$3 ($\uparrow$)

Quantity Demanded: 100 $\rightarrow$ 70 ($\downarrow$)

P $\uparrow$ and Qd $\downarrow$ $\rightarrow$ Inverse relationship

Law of Demand verified

3. SUPPLY

Definition:

Supply = Quantity of commodity firms are willing to produce and offer for sale at a given price during a specific time period

Formula:

$$Q_s = f(P)$$

Where: | Symbol | Full Form | Unit | Description | |———|———|———|———| | Qs | Quantity Supplied
| units | Amount sellers want to sell | | P | Price | Rs./unit, \$/unit | Price of the commodity |

4. LAW OF SUPPLY

Statement:

“There is a **DIRECT** relationship between price and quantity supplied, other things remaining constant”

Formula Expression:

$$\begin{cases} P \uparrow \Rightarrow Q_s \uparrow \\ P \downarrow \Rightarrow Q_s \downarrow \end{cases}$$

When to Use:

Predicting producer behavior
Analyzing supply responses to price changes
Market supply forecasting

Example:

Statement: “Wheat price increases from Rs. 20 to Rs. 25 per kg. Farmers increase supply from 500 kg to 700 kg. Verify law of supply.”

Price: Rs. 20 → Rs. 25 (↑)

Quantity Supplied: 500 → 700 (↑)

P↑ and Qs↑ → Direct relationship

Law of Supply verified

5. MARKET EQUILIBRIUM

Equilibrium Condition:

$$Q_d = Q_s$$

Where: | Symbol | Full Form | Description | |———|———|———|———| | Qd | Quantity Demanded | What consumers want to buy | | Qs | Quantity Supplied | What producers want to sell | | Pe | Equilibrium Price | Price where Qd = Qs | | Qe | Equilibrium Quantity | Quantity where Qd = Qs |

Definition:

Equilibrium = Market condition where quantity demanded equals quantity supplied - No tendency for price to change - No shortage or surplus

When to Use:

Finding market-clearing price
 Analyzing stable market conditions
 Predicting price movements

Example Problem:

Statement: “Demand: $Q_d = 100 - 2P$. Supply: $Q_s = 20 + 3P$. Find equilibrium price and quantity.”

Solution:

At equilibrium: $Q_d = Q_s$

$$100 - 2P = 20 + 3P$$

$$100 - 20 = 3P + 2P$$

$$80 = 5P$$

$$P = 16 \leftarrow \text{Equilibrium Price (Pe)}$$

$$Q_e = 100 - 2(16) = 100 - 32 = 68 \text{ units}$$

OR

$$Q_e = 20 + 3(16) = 20 + 48 = 68 \text{ units}$$

Equilibrium: $P_e = \$16$, $Q_e = 68 \text{ units}$

6. MARKET DISEQUILIBRIA

A. Excess Demand (Shortage)

Condition:

$$Q_d > Q_s \Rightarrow \text{Price } \uparrow$$

Where: | Term | Full Form | Description | |——|———|———-| | Excess Demand | Shortage | Consumers want more than available | | $Q_d > Q_s$ | Demand exceeds Supply | Not enough goods for buyers |

Market Response: Price **rises** until equilibrium restored

Example:

Statement: “At $P = \$10$, $Q_d = 100$, $Q_s = 60$. Identify the situation.”

$$Q_d = 100 > Q_s = 60$$

$$\text{Excess Demand} = 100 - 60 = 40 \text{ units (Shortage)}$$

Price will rise to eliminate shortage $\uparrow$

B. Excess Supply (Surplus)

Condition:

$$Q_s > Q_d \Rightarrow \text{Price } \downarrow$$

Where: | Term | Full Form | Description | |——|———|———| | Excess Supply | Surplus | Producers have more than consumers want | | $Q_s > Q_d$ | Supply exceeds Demand | Too many goods unsold |

Market Response: Price **falls** until equilibrium restored

Example:

Statement: “At $P = \$20$, $Q_d = 50$, $Q_s = 90$. Identify the situation.”

$Q_s = 90 > Q_d = 50$

Excess Supply = $90 - 50 = 40$ units (Surplus)

Price will fall to eliminate surplus ↓

7. CHANGES IN DEMAND

A. Movement Along Demand Curve (Change in Quantity Demanded)

Cause: Change in price of the good itself

Types:

Extension of Demand

$P \downarrow \Rightarrow Q_d \uparrow$ (move down the curve)

Example: “Price falls from \$10 to \$8, demand increases from 50 to 70 units”

Contraction of Demand

$P \uparrow \Rightarrow Q_d \downarrow$ (move up the curve)

Example: “Price rises from \$10 to \$12, demand decreases from 50 to 30 units”

B. Shift of Demand Curve (Change in Demand)

Cause: Change in factors other than own price

Demand Shifters: 1. **Consumer Income** (Y) 2. **Prices of Related Goods** (substitute/complement prices) 3. **Tastes and Preferences** 4. **Number of Consumers** 5. **Expectations** (future prices/income)

Increase in Demand

Demand Curve shifts RIGHT $\Rightarrow$ At same P, higher Q_d

Causes: - Income ↑ (for normal goods) - Price of substitute ↑ - Price of complement ↓ - Favorable change in tastes

Decrease in Demand

Demand Curve shifts LEFT $\Rightarrow$ At same P, lower Q_d

Causes: - Income ↓ (for normal goods) - Price of substitute ↓ - Price of complement ↑ - Unfavorable change in tastes

8. CHANGES IN SUPPLY

A. Movement Along Supply Curve (Change in Quantity Supplied)

Cause: Change in **price of the good itself**

Types:

Expansion of Supply

$$P \uparrow \Rightarrow Q_s \uparrow \text{ (move up the curve)}$$

Example: “Price rises from Rs. 20 to Rs. 25, supply increases from 100 to 150 units”

Contraction of Supply

$$P \downarrow \Rightarrow Q_s \downarrow \text{ (move down the curve)}$$

Example: “Price falls from Rs. 20 to Rs. 15, supply decreases from 100 to 70 units”

B. Shift of Supply Curve (Change in Supply)

Cause: Change in **factors other than own price**

Supply Shifters: 1. **Technology** (production methods) 2. **Cost of Production** (input prices) 3. **Government Policy** (taxes, subsidies) 4. **Prices of Other Goods** (alternative products firm could make) 5. **Number of Sellers** 6. **Expectations** (future prices)

Increase in Supply

$$\text{Supply Curve shifts RIGHT} \Rightarrow \text{At same P, higher } Q_s$$

Causes: - Technology improvement - Cost of production ↓ - Government subsidy - Taxes ↓

Decrease in Supply

$$\text{Supply Curve shifts LEFT} \Rightarrow \text{At same P, lower } Q_s$$

Causes: - Outdated technology - Cost of production ↑ - Taxes ↑ - Adverse weather (agriculture)

9. EFFECTS OF CHANGES ON EQUILIBRIUM

Effects of Demand Changes

Increase in Demand

$$\text{Demand} \uparrow \Rightarrow \begin{cases} P_e \uparrow \\ Q_e \uparrow \end{cases}$$

Result: Higher equilibrium price AND higher equilibrium quantity

Example: “Demand for smartphones increases” - Price ↑ - Quantity sold ↑

Decrease in Demand

$$\text{Demand } \downarrow \Rightarrow \begin{cases} P_e \downarrow \\ Q_e \downarrow \end{cases}$$

Result: Lower equilibrium price AND lower equilibrium quantity

Example: “Demand for DVDs decreases (streaming alternatives)” - Price ↓ - Quantity sold ↓

Effects of Supply Changes

Increase in Supply

$$\text{Supply } \uparrow \Rightarrow \begin{cases} P_e \downarrow \\ Q_e \uparrow \end{cases}$$

Result: Lower equilibrium price BUT higher equilibrium quantity

Example: “Good harvest increases wheat supply” - Price ↓ - Quantity available ↑

Decrease in Supply

$$\text{Supply } \downarrow \Rightarrow \begin{cases} P_e \uparrow \\ Q_e \downarrow \end{cases}$$

Result: Higher equilibrium price BUT lower equilibrium quantity

Example: “Drought reduces crop supply” - Price ↑ - Quantity available ↓

10. SUMMARY TABLE: Movement vs. Shift

Aspect	Movement	Shift
Cause	Change in own price	Change in other factors
Demand Movement	Extension (P↓) or Contraction (P↑)	Increase (right) or Decrease (left)
Supply Movement	Expansion (P↑) or Contraction (P↓)	Increase (right) or Decrease (left)
Curve	Same curve , different point	New curve
Example (Demand)	“Price falls, buy more”	“Income rises, buy more”
Example (Supply)	“Price rises, produce more”	“Technology improves, produce more”

11. TYPES OF GOODS

By Demand Response to Income

Normal Goods

$$\text{Income } \uparrow \Rightarrow \text{Demand } \uparrow$$

Examples: Cars, electronics, dining out

Inferior Goods

$$\text{Income } \uparrow \Rightarrow \text{Demand } \downarrow$$

Examples: Low-quality products, public transportation (for high earners)

By Demand Response to Related Goods

Substitutes

$$P_B \uparrow \Rightarrow Q_{d,A} \uparrow$$

Where: - P_B = Price of good B - $Q_{d,A}$ = Quantity demanded of good A

Examples: Tea & Coffee, Butter & Margarine

Complements

$$P_B \uparrow \Rightarrow Q_{d,A} \downarrow$$

Examples: Cars & Gasoline, Printers & Ink

12. QUICK REFERENCE: Key Relationships

Scenario	Demand	Supply	Price	Quantity
Price of good $\uparrow$	$Q_d \downarrow$ (movement)	$Q_s \uparrow$ (movement)	-	-
Income $\uparrow$ (normal good)	$D \uparrow$ (shift right)	-	$P_e \uparrow$	$Q_e \uparrow$
Technology $\uparrow$	-	$S \uparrow$ (shift right)	$P_e \downarrow$	$Q_e \uparrow$
Substitute price $\uparrow$	$D \uparrow$ (shift right)	-	$P_e \uparrow$	$Q_e \uparrow$
Complement price $\uparrow$	$D \downarrow$ (shift left)	-	$P_e \downarrow$	$Q_e \downarrow$
Cost of production $\uparrow$	-	$S \downarrow$ (shift left)	$P_e \uparrow$	$Q_e \downarrow$

Legend: - D = Demand curve - S = Supply curve - P_e = Equilibrium price - Q_e = Equilibrium quantity
